

ECE 447 — Equation Sheet Lessons 1–13 (Chapters 1–4)

Transform pairs and properties (Lathi & Ding Tables 3.1 and 3.2) are on page 2.

Conventions

$\text{sinc}(x) = \frac{\sin x}{x}$ — the π is written *inside* the argument, as in Table 3.1.
 $\Pi(t/\tau)$ is unity over $|t| < \tau/2$; $\Delta(t/\tau)$ has unit height and *base* τ . All powers assume normalized $R = 1 \Omega$, so $p(t) = |x(t)|^2$. B denotes message bandwidth, one-sided, in Hz. $\omega = 2\pi f$.

Signal Size

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt \quad P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

Periodic signal of period T_0 : $P_g = \frac{1}{T_0} \int_{T_0} |g(t)|^2 dt$

Energy signal: $E_g < \infty$, $P_g = 0$. Power signal: $E_g \rightarrow \infty$, $0 < P_g < \infty$.

Sinusoid: $g(t) = A \cos(\omega t + \theta) \Rightarrow P_g = \frac{A^2}{2}$

Sum of *different* frequencies: powers add, $P = \sum A_i^2/2$.

Spectral Density and Correlation

Parseval / Rayleigh:

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt = \int_{-\infty}^{\infty} |G(f)|^2 df$$

Energy spectral density and its autocorrelation:

$$\Psi_g(f) = |G(f)|^2, \quad \psi_g(\tau) = \int_{-\infty}^{\infty} g(t)g^*(t - \tau) dt$$

$$\psi_g(\tau) \longleftrightarrow \Psi_g(f), \quad E_g = \int_{-\infty}^{\infty} \Psi_g(f) df$$

Power spectral density and its autocorrelation:

$$S_g(f) = \lim_{T \rightarrow \infty} \frac{|G_T(f)|^2}{T}, \quad \mathcal{R}_g(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g(t)g^*(t - \tau) dt$$

$$\mathcal{R}_g(\tau) \longleftrightarrow S_g(f), \quad P_g = \int_{-\infty}^{\infty} S_g(f) df$$

Essential bandwidth B holding a fraction α of the energy — solve for B :

$$\int_{-B}^B \Psi_g(f) df = \alpha E_g$$

Modulating to $\pm f_c$ halves the total energy into each copy.

Trigonometric Identities and Mixing

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos^2 A = \frac{1}{2} (1 + \cos 2A) \quad \sin^2 A = \frac{1}{2} (1 - \cos 2A)$$

Multiplying two cosines makes a sum frequency and a difference frequency. A mixer followed by a filter keeps whichever one you want — that is how a signal gets translated in frequency.

LTI Systems and Distortion

$$y(t) = x(t) * h(t) \quad \longleftrightarrow \quad Y(f) = H(f)X(f)$$

Distortionless transmission: $y(t) = kx(t - t_d)$, which requires *both*, over the band the signal occupies:

$$|H(f)| = k \quad (\text{constant}) \quad \theta_h(f) = -2\pi f t_d \quad (\text{linear})$$

$$t_d = -\frac{1}{2\pi} \frac{d\theta_h}{df} \quad H(f) = k e^{-j2\pi f t_d}$$

$|H|$ not flat $\Rightarrow$ **amplitude** distortion.

θ_h not linear $\Rightarrow$ **phase (delay)** distortion.

Amplitude Modulation — DSB-SC

$$\varphi_{\text{DSB-SC}}(t) = A_c m(t) \cos \omega_c t$$

$$\Phi(f) = \frac{A_c}{2} [M(f - f_c) + M(f + f_c)]$$

$$\text{transmission bandwidth} = 2B \quad P = \frac{1}{2} A_c^2 P_m \xrightarrow{A_c=1} \frac{1}{2} P_m$$

Coherent (synchronous) detection only. With local-oscillator phase error θ , the recovered message is scaled by $\cos \theta$; $\theta = 90^\circ$ gives zero output.

Amplitude Modulation — DSB-LC

$$\varphi_{\text{AM}}(t) = [A + m(t)] \cos \omega_c t$$

$$\Phi(f) = \frac{1}{2} [M(f - f_c) + M(f + f_c)] + \frac{A}{2} [\delta(f - f_c) + \delta(f + f_c)]$$

$$\mu = \frac{m_p}{A} \quad \text{transmission bandwidth} = 2B$$

From the envelope extremes: $\mu = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$

Envelope detection valid only for $\mu \leq 1$; $\mu > 1$ is overmodulation.

$$P_c = \frac{A^2}{2}, \quad P_s = \frac{P_m}{2} = \frac{\overline{m^2(t)}}{2}, \quad P_T = P_c + P_s$$

$$\eta = \frac{P_s}{P_c + P_s} = \frac{\overline{m^2(t)}}{A^2 + \overline{m^2(t)}}$$

Single tone: $\eta = \frac{\mu^2}{\mu^2 + 2}$, max 33.3% at $\mu = 1$.

Angle Modulation

$$\varphi(t) = A \cos \theta(t), \quad \omega_i(t) = \frac{d\theta}{dt}, \quad \theta(t) = \int_{-\infty}^t \omega_i(\alpha) d\alpha$$

PM: $\theta = \omega_c t + k_p m(t)$, $\omega_i = \omega_c + k_p \dot{m}(t)$

$$\varphi_{\text{PM}}(t) = A \cos[\omega_c t + k_p m(t)], \quad \Delta f = \frac{k_p \dot{m}_p}{2\pi}$$

FM: $\theta = \omega_c t + k_f \int m d\alpha$, $\omega_i = \omega_c + k_f m(t)$

$$\varphi_{\text{FM}}(t) = A \cos \left[\omega_c t + k_f \int_{-\infty}^t m(\alpha) d\alpha \right], \quad \Delta f = \frac{k_f m_p}{2\pi}$$

Asymmetric message: $\Delta f = \frac{k_f (m_{\max} - m_{\min})}{4\pi}$

k_f is in rad/s per volt and k_p in rad per volt — that is where the $1/2\pi$ comes from. A deviation reported in hertz should never still carry a 2π .

Power is **always** $P = \frac{A^2}{2}$, independent of $m(t)$.

Deviation ratio / modulation index: $\beta = \frac{\Delta f}{B}$

Carson's rule:

$$B_{\text{FM}} = 2(\Delta f + B) = 2B(\beta + 1)$$

Narrowband $\beta \ll 1$: $B_{\text{FM}} \approx 2B$. Wideband $\beta \gg 1$: $B_{\text{FM}} \approx 2\Delta f$.

Narrowband approximations, with $a(t) = \int m d\alpha$:

$$\varphi_{\text{FM}} \approx A [\cos \omega_c t - k_f a(t) \sin \omega_c t]$$

$$\varphi_{\text{PM}} \approx A [\cos \omega_c t - k_p m(t) \sin \omega_c t]$$

FM deviation depends on message *amplitude* only; PM deviation depends on message *slope*, so it grows with message frequency.

Fourier Transform Pairs and Properties

Lathi & Ding Tables 3.1 and 3.2, in the f (hertz) convention used throughout ECE 447. Note this differs from the ω convention of the ECE 333 sheet.

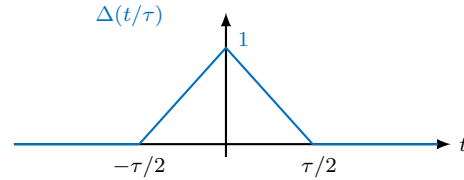
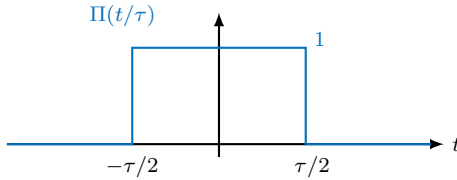
Definitions

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t} dt$$

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi f t} df$$

$$\text{sinc}(x) = \frac{\sin x}{x}$$

$$\Pi\left(\frac{t}{\tau}\right) = \begin{cases} 1, & |t| \leq \tau/2 \\ 0, & \text{otherwise} \end{cases}$$



Both shapes use *base* τ , so a triangle spanning $-T$ to $+T$ is $\Delta(t/2T)$, not $\Delta(t/T)$. Some slides write the triangle as Λ .

Transform Pairs

1. $e^{-at}u(t) \longleftrightarrow \frac{1}{a + j2\pi f}, a > 0$
2. $e^{at}u(-t) \longleftrightarrow \frac{1}{a - j2\pi f}, a > 0$
3. $e^{-a|t|} \longleftrightarrow \frac{2a}{a^2 + (2\pi f)^2}, a > 0$
4. $te^{-at}u(t) \longleftrightarrow \frac{1}{(a + j2\pi f)^2}, a > 0$
5. $t^n e^{-at}u(t) \longleftrightarrow \frac{n!}{(a + j2\pi f)^{n+1}}, a > 0$
6. $\delta(t) \longleftrightarrow 1$
7. $1 \longleftrightarrow \delta(f)$
8. $e^{j2\pi f_0 t} \longleftrightarrow \delta(f - f_0)$
9. $\cos 2\pi f_0 t \longleftrightarrow \frac{1}{2}[\delta(f + f_0) + \delta(f - f_0)]$
10. $\sin 2\pi f_0 t \longleftrightarrow j\frac{1}{2}[\delta(f + f_0) - \delta(f - f_0)]$
11. $u(t) \longleftrightarrow \frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$
12. $\text{sgn } t \longleftrightarrow \frac{2}{j2\pi f}$
13. $\cos 2\pi f_0 t u(t) \longleftrightarrow \frac{1}{4}[\delta(f - f_0) + \delta(f + f_0)] + \frac{j2\pi f}{(2\pi f_0)^2 - (2\pi f)^2}$
14. $\sin 2\pi f_0 t u(t) \longleftrightarrow \frac{1}{4j}[\delta(f - f_0) - \delta(f + f_0)] + \frac{2\pi f_0}{(2\pi f_0)^2 - (2\pi f)^2}$
15. $e^{-at} \sin 2\pi f_0 t u(t) \longleftrightarrow \frac{2\pi f_0}{(a + j2\pi f)^2 + 4\pi^2 f_0^2}, a > 0$
16. $e^{-at} \cos 2\pi f_0 t u(t) \longleftrightarrow \frac{a + j2\pi f}{(a + j2\pi f)^2 + 4\pi^2 f_0^2}, a > 0$
17. $\Pi\left(\frac{t}{\tau}\right) \longleftrightarrow \tau \text{sinc}(\pi f \tau)$
18. $2B \text{sinc}(2\pi B t) \longleftrightarrow \Pi\left(\frac{f}{2B}\right)$
19. $\Delta\left(\frac{t}{\tau}\right) \longleftrightarrow \frac{\tau}{2} \text{sinc}^2\left(\frac{\pi f \tau}{2}\right)$
20. $B \text{sinc}^2(\pi B t) \longleftrightarrow \Delta\left(\frac{f}{2B}\right)$
21. $\sum_{n=-\infty}^{\infty} \delta(t - nT) \longleftrightarrow f_0 \sum_{n=-\infty}^{\infty} \delta(f - nf_0), f_0 = \frac{1}{T}$
22. $e^{-t^2/2\sigma^2} \longleftrightarrow \sigma\sqrt{2\pi} e^{-2(\sigma\pi f)^2}$

Transform Properties

Operation	$g(t)$	$G(f)$
Linearity	$\alpha_1 g_1(t) + \alpha_2 g_2(t)$	$\alpha_1 G_1(f) + \alpha_2 G_2(f)$
Duality	$G(t)$	$g(-f)$
Time scaling	$g(at)$	$\frac{1}{ a } G\left(\frac{f}{a}\right)$
Time shifting	$g(t - t_0)$	$G(f) e^{-j2\pi f t_0}$
Frequency shifting	$g(t) e^{j2\pi f_0 t}$	$G(f - f_0)$
Time convolution	$g_1(t) * g_2(t)$	$G_1(f) G_2(f)$
Frequency convolution	$g_1(t) g_2(t)$	$G_1(f) * G_2(f)$
Time differentiation	$\frac{d^n g(t)}{dt^n}$	$(j2\pi f)^n G(f)$
Time integration	$\int_{-\infty}^t g(x) dx$	$\frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$